

Synthetic Market Phase 6

Profile-invariance checkpoint for the representation-first synthetic track. Research anchors: PHASE6_PROFILE_INVARIANCE_NOTES.md, SYNTHETIC_MARKET_REPRESENTATION_CONTROLS.typ, and RANKED_RESEARCH_ROADMAP.md.

DATE	PROMPTS	DESIGN	LAYERS
21 March 2026	48 market-only	2 scenarios × 4 styles × 6 layouts	48

MAIN READ

Phase 6 says the abstraction bottleneck is layout, not wording.

Primitive market factors remain almost perfectly explicit in row states. Under the harder profile-invariance control, participation × concentration still survives better than momentum × flow. But the decomposition shows the crucial point: changing surface style barely breaks retrieval, while moving profiles across row layouts does.

BEST PRIMITIVE R ²	BEST FULL CONTROL	BEST STYLE-ONLY	BEST LAYOUT-ONLY
0.9998 attractiveness, row_mean L1	0.032 part./conc., row_eos L16	0.121 part./conc., row_eos L15	0.011 part./conc., row_eos L21

Scope

Phase 6 is not another generic synthetic probe run. It isolates one very specific question:

- if profile-level market abstractions exist, what kinds of nuisance changes actually break them?

The dataset keeps the latent profiles fixed while changing:

- surface style
- display symbols
- row layout

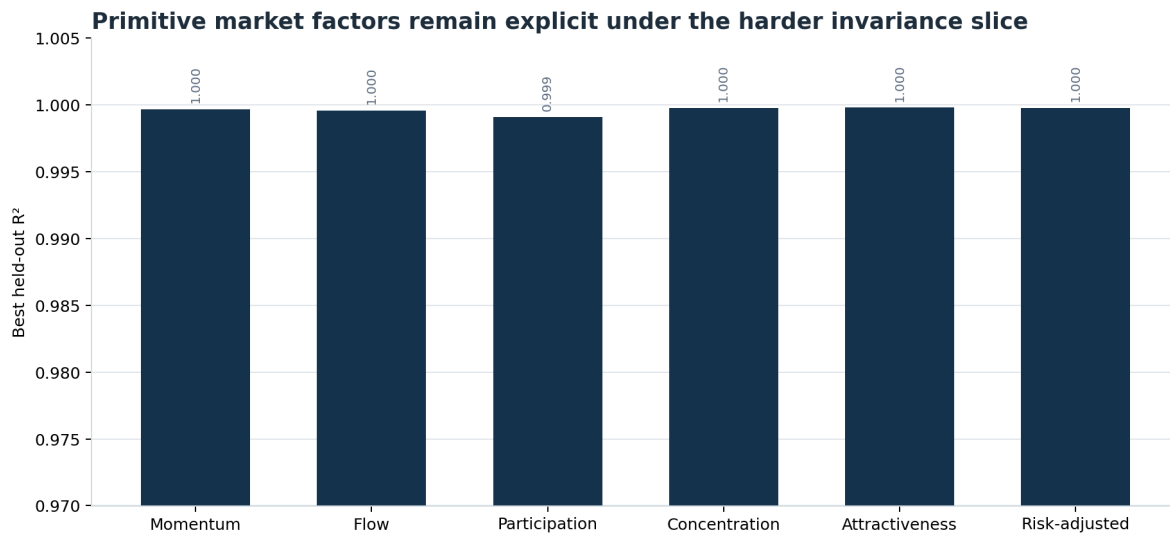
This phase covers two scenario families:

- momentum_flow_tiebreak
- participation_concentration_tiebreak

The important methodological shift is that Phase 6 does **not** treat all nuisance factors as one undifferentiated stress test. The report decomposes the failure into **style-only** and **layout-only** retrieval.

Primitive Factors Still Hold

The harder invariance slice does not challenge primitive-factor representation.



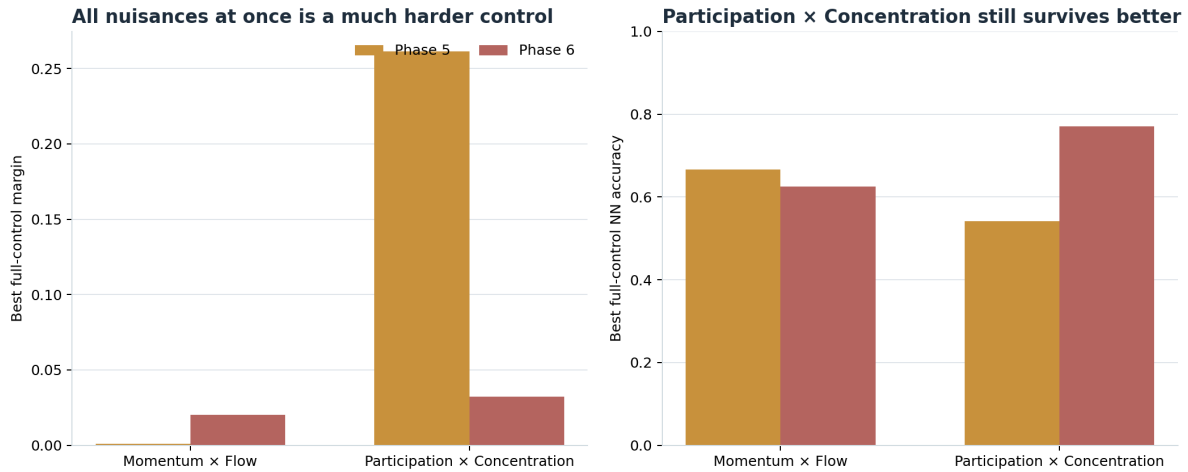
Best held-out primitive regression performance from Phase 6. Primitive market variables remain almost perfectly recoverable from row_mean, usually by layer 1.

Variable	Best State	Metric	Layer
pct_5m	row_mean	R ² 0.9997	L1
net_flow_5m	row_mean	R ² 0.9996	L1
unique_traders_5m	row_mean	R ² 0.9991	L1
top20_holder_pct	row_mean	R ² 0.9998	L1
attractiveness_score	row_mean	R ² 0.9998	L1
risk_adjusted_score	row_mean	R ² 0.9998	L1

This remains the most stable representation claim in the project: the row states explicitly carry primitive market factors. The interesting uncertainty is higher-order abstraction, not raw factor storage.

Full Control

Compared to Phase 5, the all-nuisance Phase 6 control is much harsher.



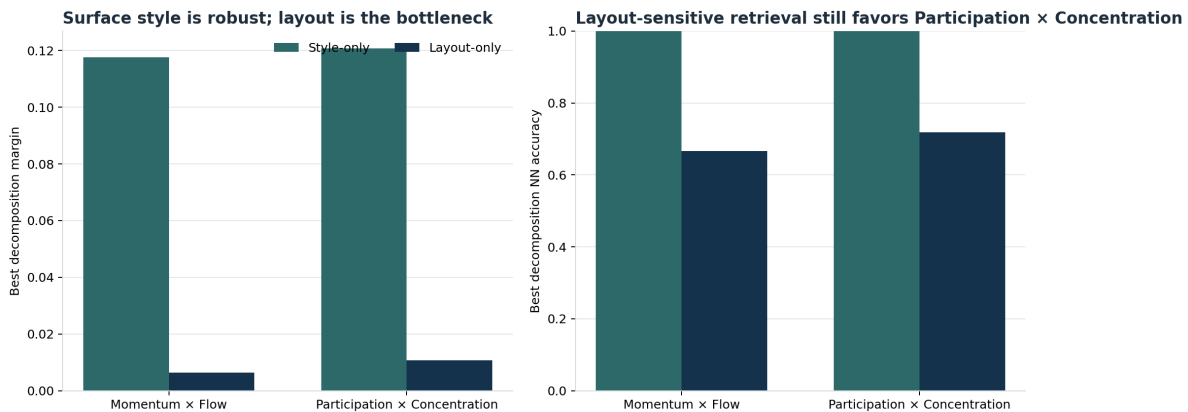
Phase 5 versus Phase 6 on the strict full profile-control metric. The harder combined nuisance stack compresses both families, but participation/concentration still holds up better.

Scenario	Margin	NN Acc.	Read
momentum x flow	0.20	0.625	Still above zero, but only weakly separated under the full nuisance stack.
participation x concentration	0.32	0.771	Remains the stronger profile-level candidate under the hardest read.

This is enough to keep participation x concentration as the lead abstraction family. It is **not** enough to claim a broadly nuisance-invariant market profile representation.

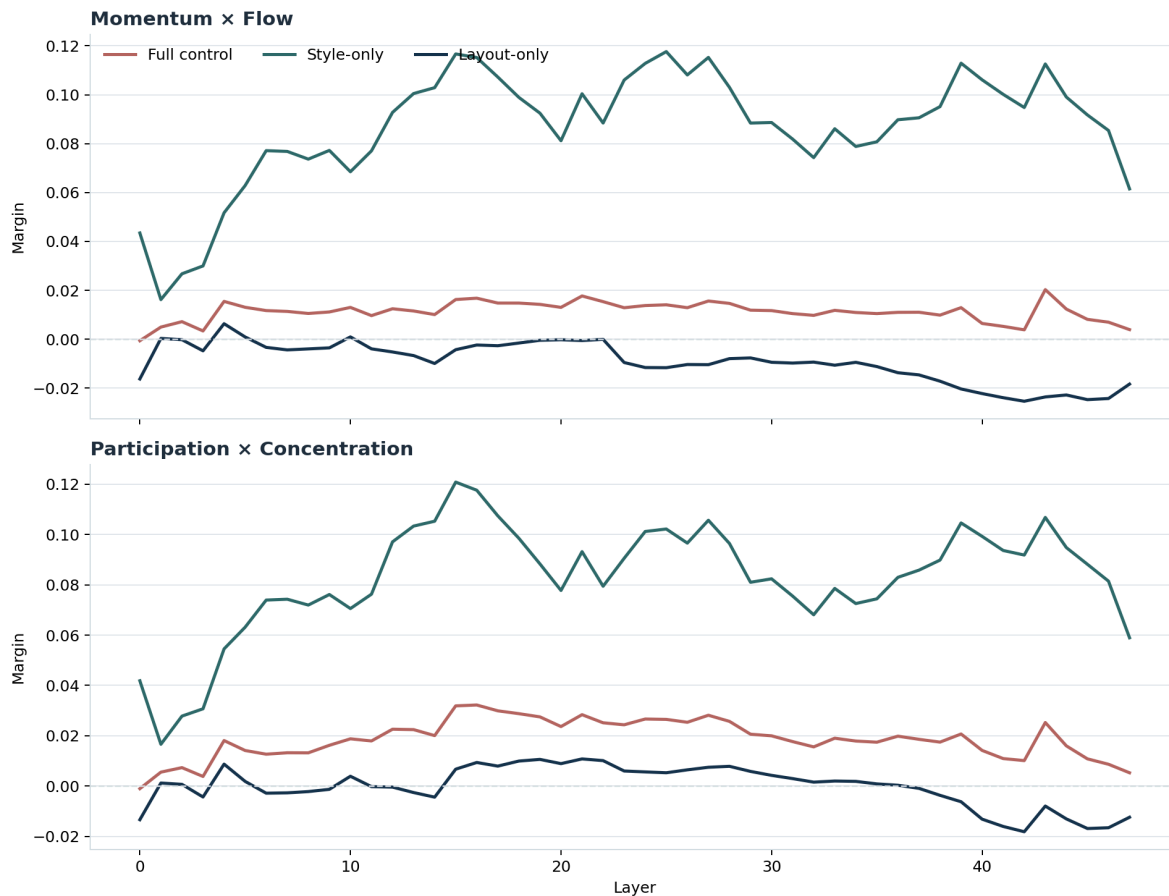
What Actually Breaks

The decomposition is the main new result.



Best decomposition metrics by scenario. Style-only retrieval is strong for both families. Layout-only retrieval is the real bottleneck, and participation/concentration remains better on that harder axis.

Decomposing the nuisance stack shows what actually breaks invariance



Layerwise margins for full control, style-only retrieval, and layout-only retrieval. The gap between style-only and layout-only is the core Phase 6 finding.

Scenario	Mode	Best Margin	Best NN Acc.
momentum × flow	style-only	0.118	1.000
momentum × flow	layout-only	0.006	0.667
participation × concentration	style-only	0.121	1.000
participation × concentration	layout-only	0.011	0.719

Interpretation:

- surface wording / formatting / alias changes are **not** the main problem
- layout movement is the main problem
- participation × concentration still has the better layout-sensitive signal
- the next useful experiment should therefore target **row-layout invariance**, not more paraphrase-only variation

Why This Matters

SUPPORTED NOW

Primitive market factors are explicit in row states, and profile retrieval is substantially more robust to style changes than to row-layout changes.

NOT SUPPORTED YET

We still do **not** have a strongly row-layout-invariant profile abstraction. The best full-control and layout-only margins are positive, but still small.

This changes the next step. The project should not spend its next iteration on more generic manifold plots or more paraphrase variants. The right next move is a layout-targeted representation study:

- keep surface style fixed
- vary roster position and distractor composition harder
- test whether pairwise-relative or difference-vector representations recover profile identity better than raw row retrieval

That is the cleanest way to keep the work upstream and representation-focused without repeating the same test again.